# Week 11 web page: delivery guide

Instructor notes for Week11_LinearModelsAndVariance.html

## General

- Page: https://kevintshoemaker.github.io/NRES-746/Week11_LinearModelsAndVariance.html. Open it before class; the widget needs no internet once loaded.
- For live demos, run the matching section of `CODE_Week11_LinearModelsAndVariance.R` (same seeds, same numbers). Run its first section before any later one; everything uses `chron`, `rings`, and `tree_means`.
- The pinyon data are simulated (seed 21). Truth: stand SD 0.12, tree SD 0.10, precipitation slope 0.025 mm/cm, AR(1) $\rho = 0.6$ within stands (innovation SD 0.08), **no trend**. Say up front that they're simulated.
- Questions on the page have no answers. *Land:* is where discussion should end up.
- WODs: *Linear or Not?* (Monday) and *Two Cores from One Tree* (Wednesday); keys in `wod/answer_keys/`. No take-home WOD this week.
- Veterans Day is Wed., Nov. 11, so next week has Monday and Lab 10 only.

## Part 1: Standard statistics as likelihood models (Monday, ~32 min)

### Big picture (~3 min)

- MODELS box. Every intro-stats procedure is a data-generating model we could write ourselves; `lm()` and `glm()` are fast special cases.
- Name the independence assumption now; Wednesday relaxes it.

### WOD: linear or not? (~10 min real)

- Run the WOD before scrolling to section 2. Then read the three bullets.
- If someone says (e) can't be a GLM: it can, as a Normal GLM with an inverse link ($1/\mu = 1/a + (b/a)(1/x)$). Fine to mention, not to dwell on.

### One model behind lm() (~10 min)

- Show the two figures. Point out that the stands differ in level and share a precipitation response.

- `optim()` and `lm(width ~ 0 + stand + precip)` agree to 6 decimals; $\sigma$: MLE 0.111 vs. `lm()` 0.113 ($n$ vs. $n-5$).
- "Each precipitation value is shared by all four stands in a year. Does that matter for the independence assumption?" *Land:* no, by itself. Independence is about the residuals given the predictors. Precipitation is a predictor, so sharing it is fine. What would matter is stands sharing an unmeasured year effect (a residual).

**The model matrix (~10 min)**

- Compare the two codings on the 1996 rows; same fitted values, different meaning. standB $= -0.27$ = B's intercept minus A's.
- "In a model with an intercept, why can't R include indicator columns for all four stands?" *Land:* the four indicators add up to the intercept column, so the coefficients aren't identifiable (infinitely many solutions). Sets up the lab's $X^TX$ question.
- Don't compute $X^TX$ diagonals on the board (Lab 9 1c).

**GLMs (~7 min)**

- Table, then the cone example: `optim()` and `glm()` both give slope 3.13 on the log scale.
- One sentence on IRLS; one on when you still need a custom likelihood.
- **Behind?** Skip the GLM code (just the table) and the `nls()` bonus.

**Discussion (~5 min)**

- Q1 (residual lag-1 correlation 0.45; would you trust `lm()`'s SE for a trend?) *Land:* breaks independence. No: positively autocorrelated residuals make `lm()`'s SE too small, especially for slow predictors like year. Wednesday shows it with stand A.
- Q2 (predicting a fifth stand) *Land:* the fixed-stand model has no intercept for an unsampled stand. You'd need to treat stands as a sample from a population of stands: a random effect.

# Part 2: Linear algebra for statistics (Lab 9 pre-lab, ~30 min)

- Board: multiply a $2 \times 3$ by a $3 \times 1$ by hand, row by column, then show `%*%`. Do one row of $X\beta$ (stand C, 30 cm).
- $2 \times 2$ inverse formula on the board; check it with `solve()`.
- Normal equations: state, don't derive. Same estimates as `lm()`.
- "Suppose $X$ also had an indicator column for stand A. What would go wrong when we call `solve(XtX)`?" *Land:* $X^TX$ would be singular (determinant 0, an eigenvalue of 0), so `solve()` fails. Same point as Monday's question, now in matrix form.
- MVN: walk the three analogies to the univariate normal. The `dmvnorm()` vs. `sum(dnorm())` check gives 78.32 both ways.
- Positive definite: eigenvalue check on a different matrix (variances 4 and 2). Don't use Bolker's 0.9/0.9/−0.9 example (Lab 9 1f).

- **Don't give away:** the meaning of $X^TX$'s diagonal (1c); which residual pattern is more surprising under correlation (1e); why the $3 \times 3$ matrix fails (1f); what `mvrnorm()` does with it (2c).
- **Lab 9** (one Act, ~2 h): Part 1 paper ~40 min, group check-in ~15, laptops ~50, closing ~15. The closing block-diagonal $V$ sets up Wednesday; let groups say what it means without confirming the random-intercept vocabulary. Data: `data/PinyonSeedlings.csv` (24 seedlings, seed 4; truth: plot effects 0, 1.5, −1; slope 0.5; SD 0.8, 0.8, 1.6). Key: `labs/answer_keys/Lab9_Act1_Key.pdf`.

# Part 3: Modeling variance (Wednesday, ~33 min)

### Big picture (~3 min)

- MODELS box, noise part. Every NLL so far ends in `-sum(...)` because of independence. Correlated data need one joint density: the MVN with a structured $V$. Name what's ahead: mixed models, time series, spatial.

### WOD: two cores (~10 min real)

- Run it, then the page's general formula and the $n_{\text{eff}}$ plot. Answers: 0.02 and 0.032; Design A 0.0328 vs. B 0.004; ten same-tree cores are worth about 1.2 independent ones.
- Point at the curve for 0.8 flattening near $1/\rho = 1.25$.

### Random intercept (~10 min)

- Derive the three covariance facts on the board from the WOD formula; the page has them for reference. Connect to the lab's closing matrix.
- Fit: $\hat{\sigma}_b = 0.14$, $\hat{\sigma} = 0.085$, ICC 0.74; `lmer()` matches with `REML = FALSE`.
- "The variance among stands is estimated from only four stands. How much would you trust $\hat{\sigma}_b = 0.14$?" *Land:* not much; four draws of $b_j$. Preview the surface figure: plausible range about 0.08 to 0.35.

### AR(1) (~10 min)

- Management question first: is growth declining? `lm()`: trend −0.0055 mm/yr, SE 0.0026, $p = 0.045$. Residual ACF shows lag-1 correlation.
- Derive $\text{Var}(e_t) = \sigma^2/(1-\rho^2)$ on the board (two lines). AR(1) fit: $\hat{\rho} = 0.37$; trend −0.0047, SE 0.0034, $p = 0.17$. `gls()` matches. Then tell them the truth has no trend.
- Widget walkthrough (`cov_structures.html`):
  - Random intercept default ($\sigma_b$ 0.12, $\sigma$ 0.10): ICC 0.59, block heatmap.
  - Slide $\sigma_b$ to 0: heatmap goes diagonal, groups look alike. Then $\sigma_b$ 0.3, $\sigma$ 0.05: ICC 0.97, points clump by group. Click "New random draw" a few times.
  - Switch to AR(1), default $\rho$ 0.6: lag-1 0.60, lag-5 0.08.
  - Ask students to predict the series at $\rho = 0.9$ before moving the slider. It wanders like a trend (lag-5 still 0.59).
  - $\rho = -0.7$: zigzag series, checkerboard heatmap.

- "With $\rho$ near 0.9, what does a single AR(1) series look like? How could you tell it apart from a slow trend?" *Land:* long runs above and below the mean that look like trends. From one short series you often can't; you need a longer record, replicate series, or a mechanism. This is the trend-vs.-$\rho$ fan in section 5.

**Why harder (~8 min)**

- Three points, one line each: cost (here about 170 ms per MVN evaluation at $n = 2000$; timing depends on the machine), awkward surfaces, constraints.
- Random-intercept surface: $\sigma_b$ range about 0.08 to 0.35; $\sigma$ much tighter.
- Trend-vs.-$\rho$ fan: near $\rho = 0$ the trend runs about $-0.0074$ to $-0.0039$ (excludes 0); at $\hat{\rho} = 0.37$ it runs $-0.0113$ to 0.0017. A mean parameter's conclusion depends on a variance parameter.
- **Behind?** Skip the timing; show the two surfaces.

**Discussion (~5 min)**

- Q1 (trend SE 0.0026 to 0.0034; precipitation SE 0.0026 to 0.0020) *Land:* year is itself perfectly autocorrelated, so year-to-year error runs line up with it and mimic a trend; precipitation jumps around, so correlated errors don't line up with it. Errors matter most for predictors that change slowly in time or space.
- Q2 (four stands: fixed or random?) *Land:* the random-intercept model answers "how variable are stands in general" and can predict a new stand, but $\sigma_b$ from four stands is weak (wide surface). The fixed model answers "how do these four stands differ." Choose by the question; with few groups, say so and report the uncertainty.

**Bonuses** (only if time): combined random intercept + AR(1) ($\hat{\rho} = 0.51$, matches `lme()` to the log-likelihood, 82.28); integrating out a random effect leads into next week's Bayesian multilevel models.